

STATUS: WARNING - INVESTIGATING

INC-2839 - TWR-LAG-W-031 AI-Predicted Thermal Failure

Date: 2026-05-01

Severity: WARNING

Status: INVESTIGATING

1. Executive Summary

TelcoPilot AI model predicts an 87% probability of hardware fault by 18:42 WAT today. The system has flagged this as a high-risk event requiring urgent preventive action.

2. AI Prediction Detail

Thermal sensor readings have climbed steadily from 58C to 71C over the past 14 days. Peak load exceeded 82%, correlating with rising thermal stress. AI model projects imminent amplifier failure within the next 2 hours.

3. Thermal Trend Table: 14-Day Daily Readings

Day	Temperature (C)	Load (%)
1	58	62
2	59	63
3	60	64
4	61	65
5	62	66
6	63	67
7	64	68
8	65	69
9	66	70
10	67	71
11	68	72
12	69	73
13	70	74
14	71	75

4. Current Metrics

Metric	Value
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Signal	64%
Load	74%
Status	WARN
Issue	Predicted failure in 2h

5. Recommended Actions

Priority	Action	Owner
URGENT	Schedule maintenance window in next 2 hours	NOC Ops
URGENT	Pre-provision spare amplifier for hot-swap	Field Eng
MONITOR	Subscribe NOC to thermal-trend alerts at 80% threshold	NOC Ops

6. Cost Comparison

Scenario	Estimated Cost (NGN)	Notes
Preventive Maintenance	1.2M	Includes labor + spare amplifier
Reactive Repair	4.8M	Includes downtime, emergency dispatch, service disruption

7. Historical Reference

A similar thermal escalation pattern was observed and resolved by amplifier swap on TWR-IKJ-019 (Ikeja Allen) in March 2025. Preventive replacement avoided extended downtime.

8. Sign-off

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